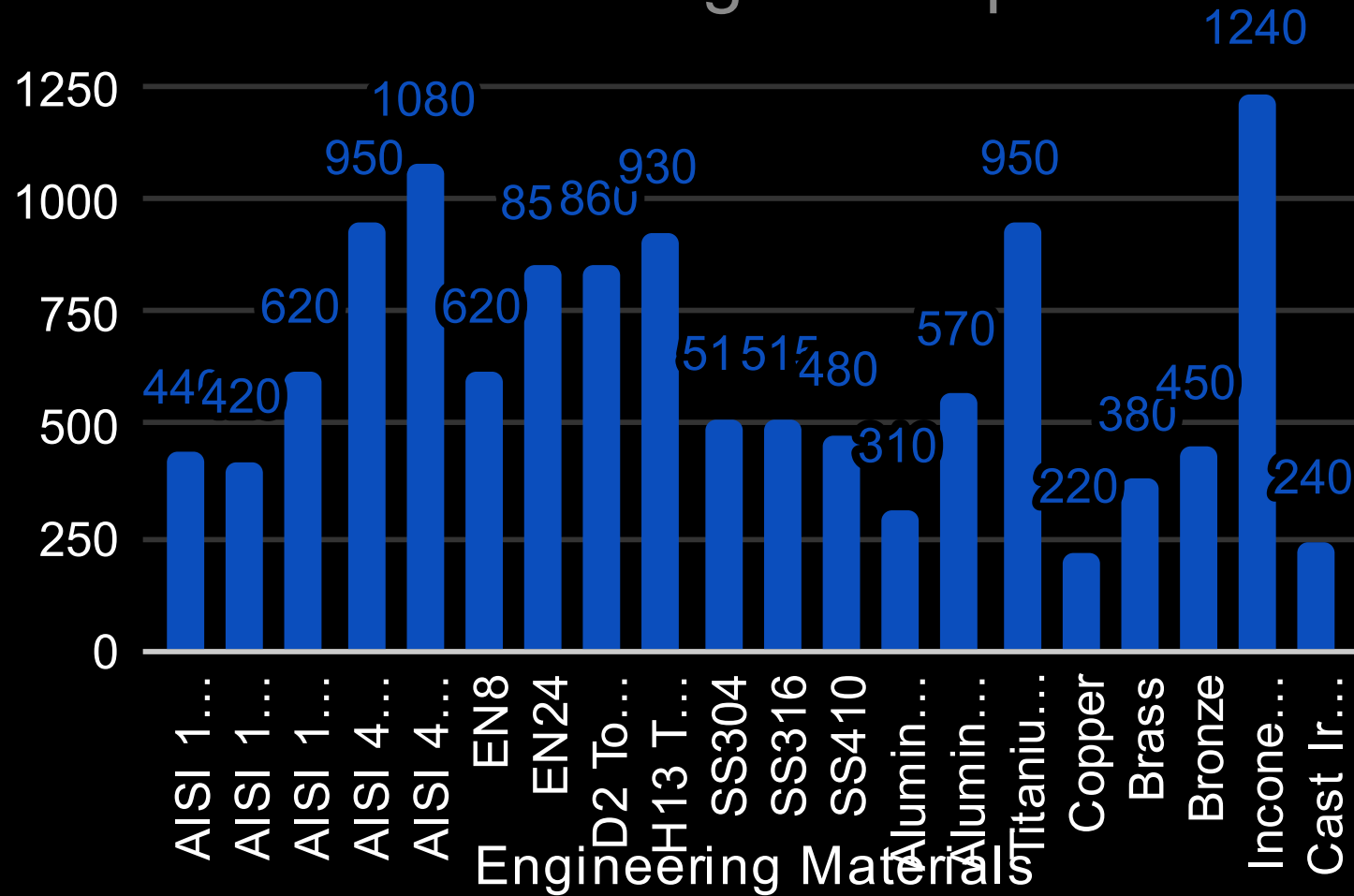


Material	Density (g/cm³)	UTS (MPa)	Yield Strength (MPa)	Hardness (HB)	Young's Modulus (GPa)
AISI 1018	7.87	440	370	126	205
AISI 1020	7.87	420	350	120	205
AISI 1045	7.85	620	530	170	210
AISI 4140	7.85	950	655	285	210
AISI 4340	7.83	1080	745	320	210
EN8	7.85	620	530	180	210
EN24	7.83	850	680	248	210
D2 Tool Steel	7.7	860	700	620	210
H13 Tool Steel	7.8	930	760	510	210
SS304	8	515	205	201	193
SS316	8	515	205	217	193
SS410	7.75	480	275	190	200
Aluminium 6061	2.7	310	276	95	69
Aluminium 7075	2.81	570	505	150	72
Titanium Grade	4.43	950	880	349	114
Copper	8.96	220	70	45	117
Brass	8.5	380	200	100	100
Bronze	8.8	450	240	120	110
Inconel 718	8.19	1240	1035	363	200
Cast Iron	7.2	240	130	210	170

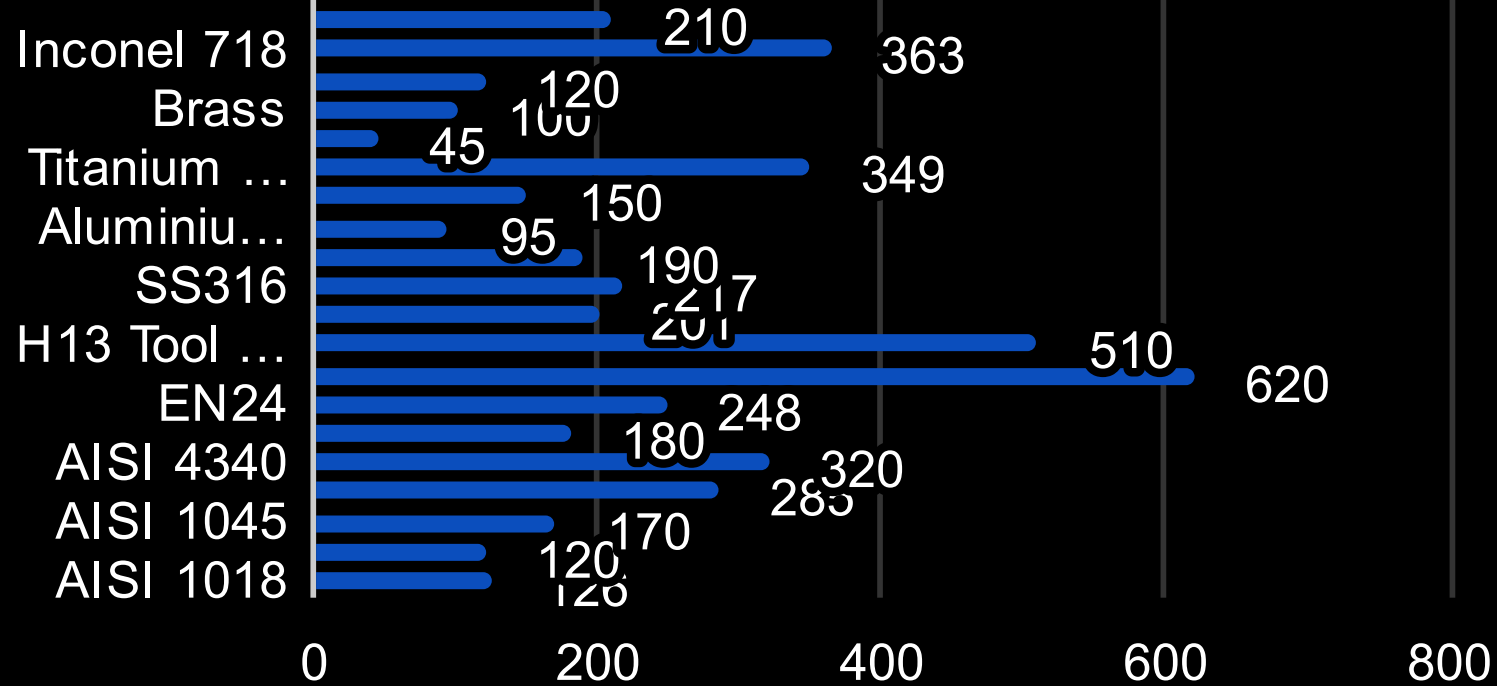
Ultimate Tensile Strength Comparison

Ultimate Tensile Strength (MPa)



Brinell Hardness Comparison

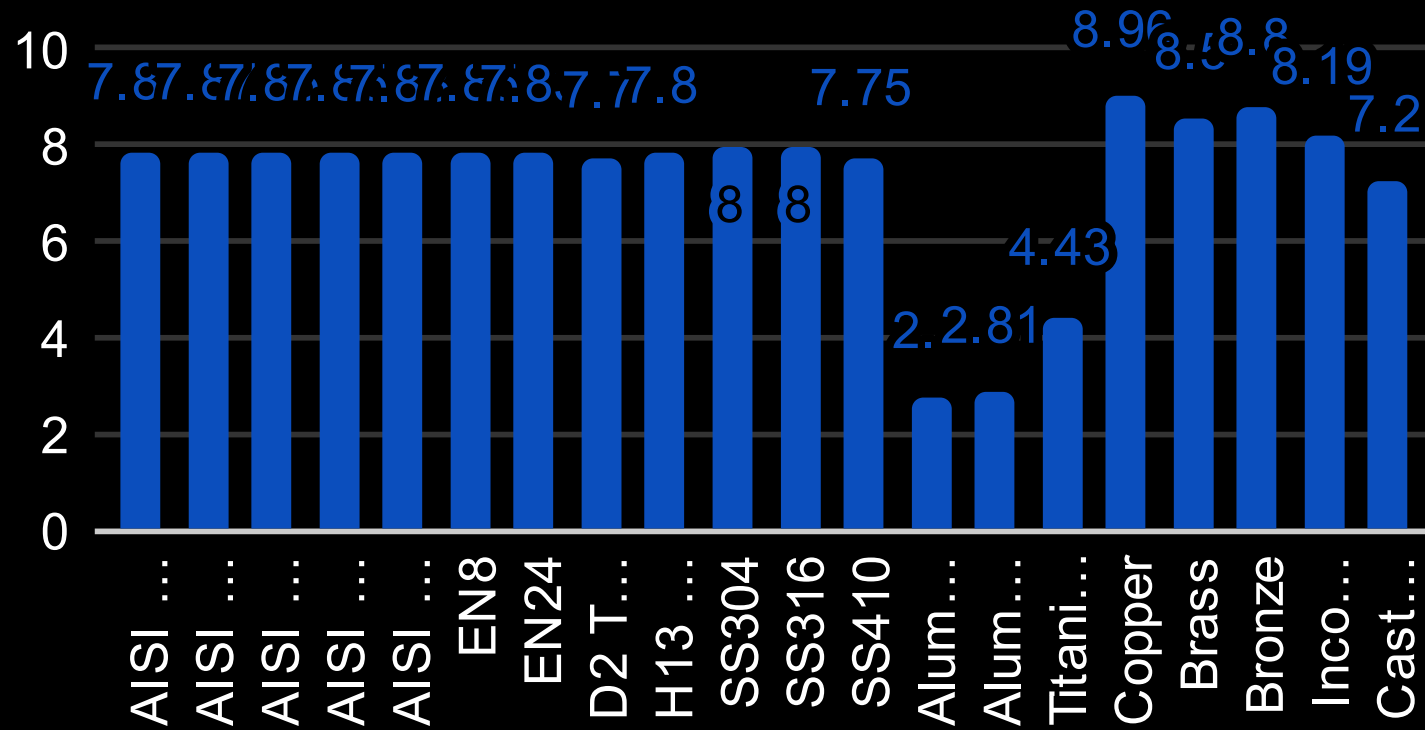
Engineering Materials



Hardness Values in BHN

Density Comparison

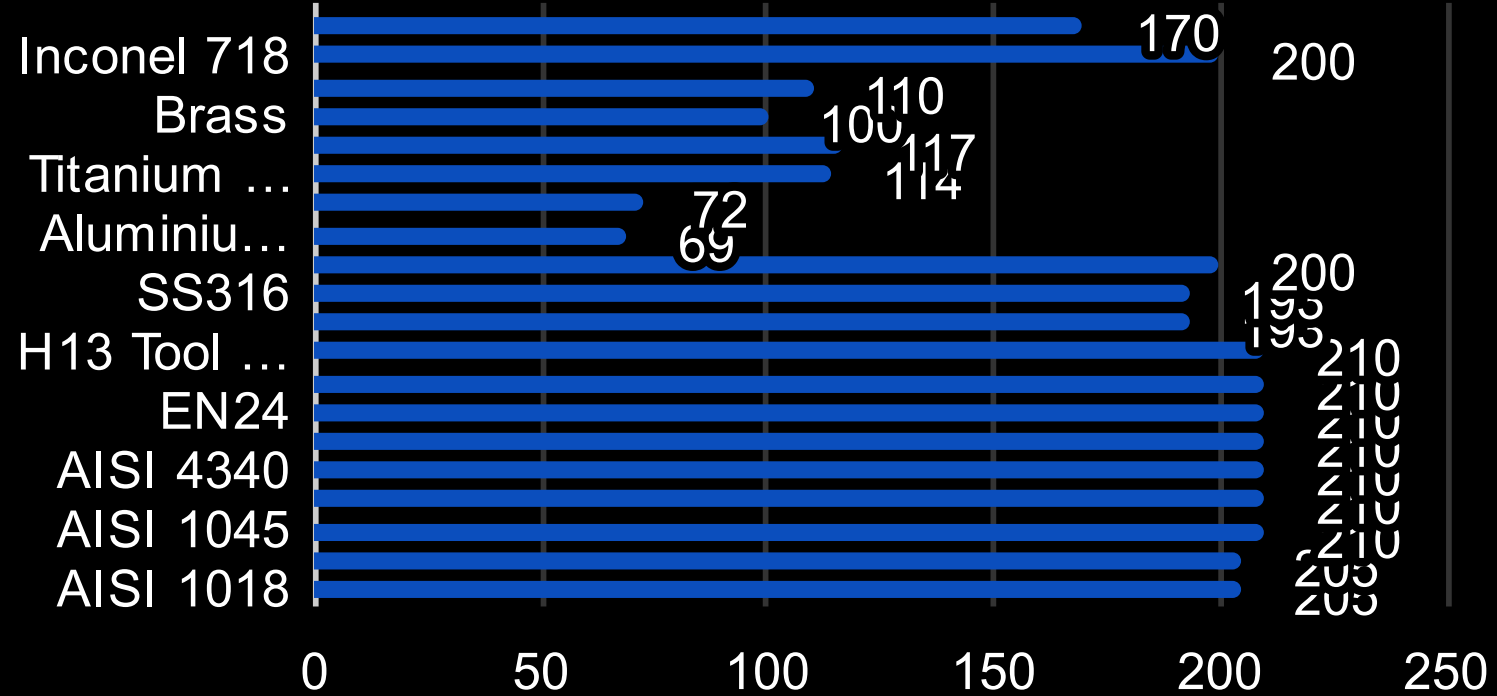
Densities



Engineering Materials

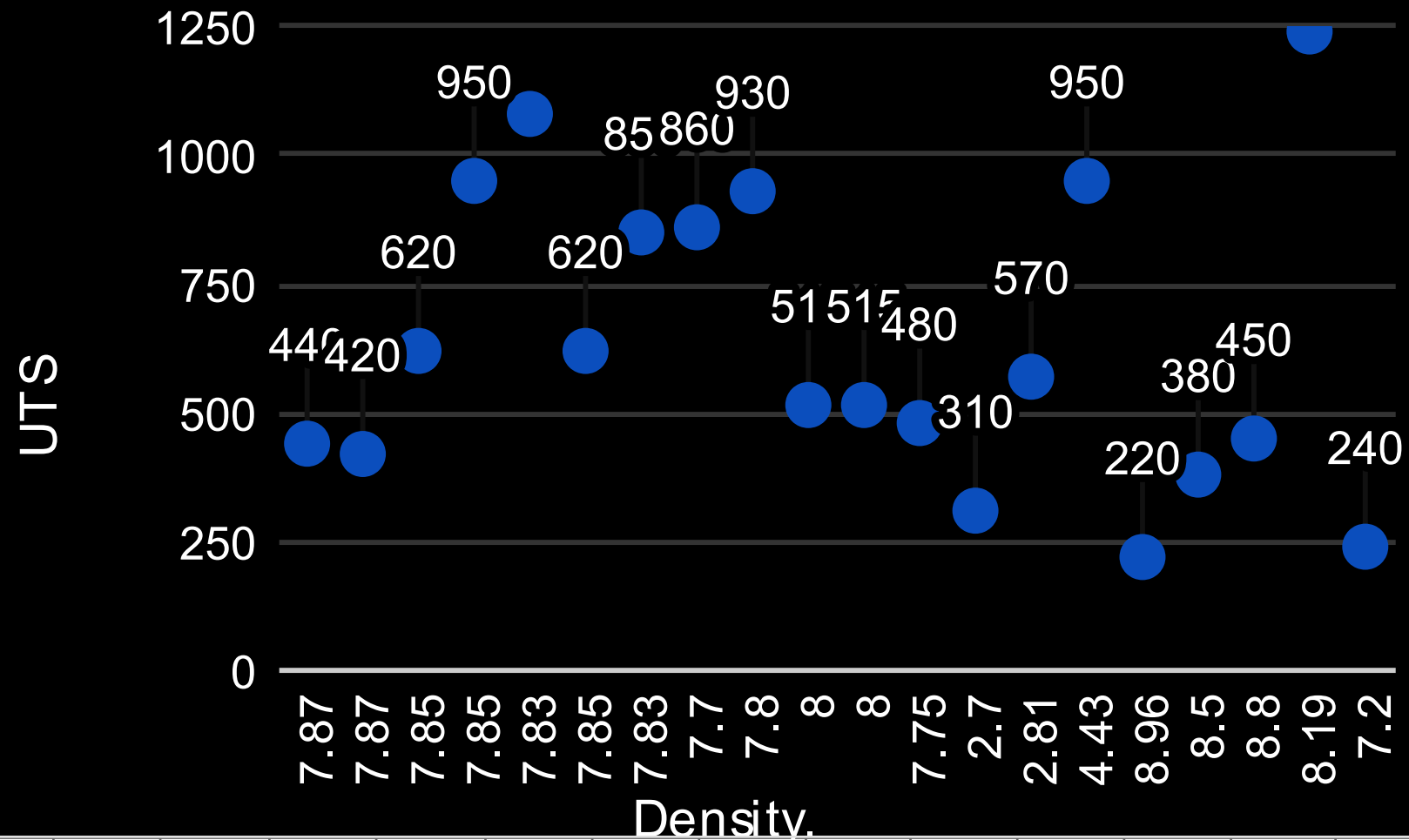
Young's Modulus Comparison

Engineering Materials



Young's Modulus

Density vs Ultimate Tensile Strength



ENGINEERING MATERIALS DATA ANALYSIS DASHBOARD



HIGHEST UTS
1240 MPa
Inconel 718



HIGHEST HARDNESS
620 HB
D2 Tool Steel



LOWEST DENSITY
2.70 g/cm³
Aluminium 6061

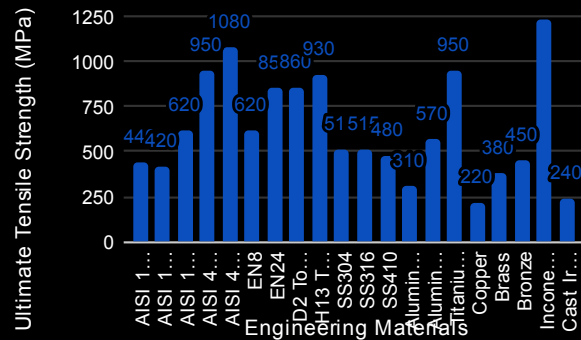


TOTAL MATERIALS
20
Materials Analyzed

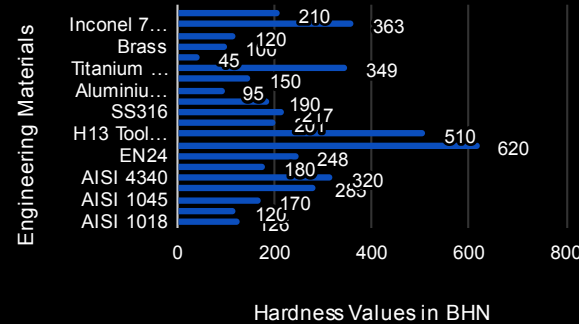
KEY INSIGHTS

- Highest tensile strength: Inconel 718 (1240 MPa)
- Highest hardness: D2 Tool Steel (620 HB)
- Lowest density: Aluminium 6061 (2.70 g/cm³)
- Titanium Grade 5 offers an excellent strength-to-weight ratio.
- Material selection should balance strength, weight, hardness, and application requirements.

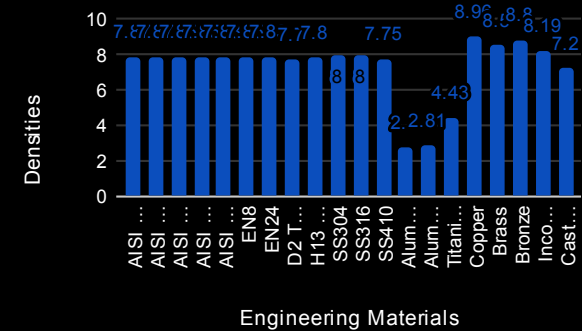
Ultimate Tensile Strength Comparison



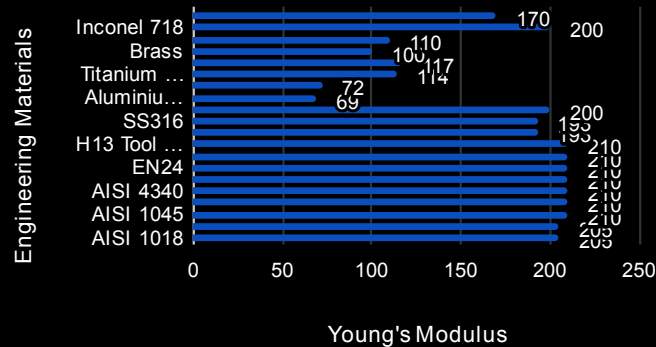
Brinell Hardness Comparison



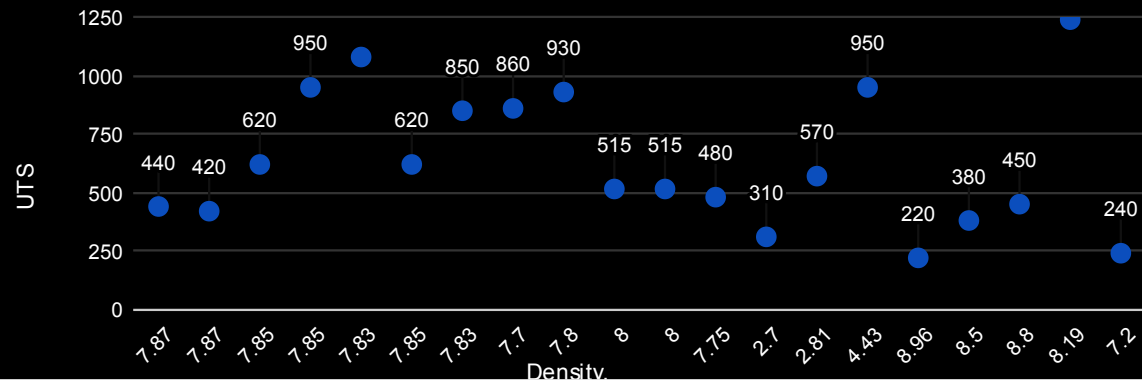
Density Comparison



Young's Modulus Comparison



Density vs Ultimate Tensile Strength



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